

Key Papers Summary

Gone with the clouds: estimating the electricity and water footprint of digital data services in Europe - Javier Farfan, Alena Lohrmann (2023)

Abstract: Publicly available data was combined with literature research to extrapolate the development of digital data usage in Europe until 2030. Results demonstrate that 273.4 to 820.1 million cubic meters of water and from 56.3 to 169 Terawatt-hour of electricity will be consumed yearly in Europe in 2030 for internet usage.

- Large computing centers still consume the electricity equivalent to that of small cities
- The immense amount of heat generated by the computing centers requires additional on-side water cooling, further increasing the water footprint of computing centers
- There is a challenge of lack of transparency on behalf of the main providers of computing services: Google, Meta, Microsoft and Amazon. Make it difficult for governments and decision makers to deal with electricity and water management in the zones where computing centers operate or may be commissioned.
- Optimization for energy and water use can be a trade-off in some cases reducing the processing performance of datacenters. This issue is particularly relevant in areas of high water stress.
- This research aims to estimate and make a projection of the yearly electricity and water consumption incurred by the currently operating and planned computing centers and data transmission in Europe until 2030

Estimation of water demand of data centers:

- Data centers are linked to high water - Water consumption by data centers is a constantly developing field
- (Indirect water consumption component) Depending on the electricity source, the water consumption for electricity production can vary greatly, where thermal power plants and hydropower plants have considerably higher water consumption than renewables like solar photovoltaic and wind.
- (Direct water consumption component) used for cooling of the data center. This is more controversial even though less consuming - as new data centers open in areas of high water stress, with the typical data
- Estimation that 1 Megawatt data center uses 25.5 million liters of water per year for cooling
- Water consumption takes place in relation to energy use rather than capacity - it was assumed that a 1 MW data center that operated without interruption consumes 8670 megawatt-hours of electricity per year.

Formulas used:

Results:

Discussion and Conclusion:

- The lack of transparency by data service providers makes it challenging to place a struffy such as this in a broader context.
- Because of the “double water impact” of data usage, the impact in water is comparatively more significant than that of the electricity.

- Moreover, the water impact from the energy component is directly influenced by the composition of the energy system. Therefore, the overall water impact of data usage can be dramatically reduced through transitioning the energy system towards renewables
- Not all digital data communication results in a detriment to the environment. For example, the digitalization of controls and sensors can help optimize resource management systems, potentially reducing the environmental impact of water and energy systems alike.

Stats facts:

- It is estimated that the average European citizen uses around 187.3 GB of data yearly in 2020 numbers. This represents an increase of 32.4% from the previous year and 286% increase compared to 5 years ago.
- Energy efficiency of data centers is constantly increasing to the point that an increase in computing workload of 550% from 2010 to 2018 has resulted only in an increase of energy demand of 6%.
- The water consumption of the average European inhabitant is 128 L of water per day, which adds up to 46.7 cubic meters of water per year.
- If the estimated average of 1.1 cubic meters per capita per year becomes a reality, the daily water consumption for data usage becomes around 3 liters per day, which would mean that the average European would be using more water in internet than the amount needed for drinking

Future work:

- The impact of data usage in the larger context regarding energy and water is yet to be thoroughly investigated.
- Although these findings are informative in nature, the awareness on the water and energy consumption of digital data services in Europe could function as a reference for future digital infrastructure developments. However, one possible application of this knowledge is for decision-making on where the next data centers should be deployed (logically, locations with the shortest distance to large population centers but in areas of low water stress)

Datasets used:

- reported development of the subscription and data usage behaviours in the OECD-Europe from 2016 to 2020
- Distribution of direct and indirect components of water consumption for data usage for all OECD-Europe countries

Wattnet: matching electricity consumption with low-carbon, low-water footprint energy supply (2026)

Abstract:

- Water-related impacts are often overlooked despite the strong interdependence between energy and water systems → particularly relevant for electricity-intensive activities such as data center operations where both carbon emissions and water use occur largely off-site through electricity consumption
- Wattnet assesses the CF and WF of electricity consumption across Europe.
- Neglecting electricity flows and temporal variability can lead to significant misestimations of both CF and WF particularly in countries with high levels of electricity trade or hydropower dependence.
- Wattnet facilitated informed decision making for workload scheduling and energy-aware operation of DCs while also enhancing transparency regarding the environmental impacts of electricity consumption for end users and policymakers.
- CF and DC has been the focus of numerous research works, including the estimation of individual workload executions.
- Most existing assessments either neglect cross-border electricity flows or focus exclusively on carbon emissions, overlooking the coupled water impacts of electricity generation
- The use of water for cooling has the advantage of reducing electricity consumption but it may also impose other environmental changes, especially in regions where water is scarce
- CF and WF of electricity consumption should not be analysed independently
- On-site water consumption can be monitored and reported but this is still not a widespread practice
- In contrast to power consumption - which only occurs on-site, the part of water consumption that takes place off-site is not as easy to compute.
 - The water used off-site in the production of energy depends on the type of technology for electricity generation - its assessment requires having knowledge about the energy mix in the grid that supplies the DC with electricity
- Regions continuously interchange energy with neighbours in the form of bidirectional fluxes with only one direction active at a time . Creates a dynamic balance where some regions are net exporters and others are net importers.

Methods:

- The GHG Protocol is the standard that defines how organizations can categorize their GHG emissions - the protocol establishes how to calculate the CF of electricity consumption
- Implementation of a flow tracing methodology to calculate each grid instant mix and ending with the calculation of CF and WF using operational or life-cycle factors

Data:

- Data from electricity generation and cross-border physical flows are obtained from the **European Network of Transmission System Operators for Electricity (ENTSO-E transparency platform)**

- Actual Generation per production type
- Cross-Border Physical flows

Electricity flow tracing:

- The electricity consumed in a given European electric grid is composed not only of the electricity generated in the region where this grid is located, but also of the electricity that this region imports and gets mixed with its own generation.
- The flow-tracing methodology allows the assessment of contributions of individual generators to individual line flows - it can be used to calculate the energy mix composition of a country by considering the country of origin of its electricity and the fluxes between neighbouring countries.
- The highest level of granularity is achieved by considering as nodes the smallest geographical areas available

Formulas:

CF and WF:

- The energy from each generator node k that arrives at each node i has a CF and WF that depends on the instant generation mix in that node k or region.
- The footprint of the (composition) energy can be calculated by iterating the footprints of the region of origin in the proportion that they contribute
- To calculate the CF and WF of the electricity produced in each generator node k , the amount of electricity produced per type of technology is multiplied by the carbon emission and water consumption factors for each technology.
- The footprint can be calculated considering the whole life cycle of the consumed energy (including infrastructure building, transportation, maintenance, waste disposal etc.) instead of only considering the direct impact of electricity generation.

Results:

- Local and Global Carbon Footprint:
 - The term local footprint is employed to denote the footprint calculated using only locally generated electricity
 - The term global footprint is used to denote the footprint that accounts for electricity imports and exports (or consumption-based)
 - Results go more in detail about country specific footprint (interesting) and why
- Local and Global Water Footprint:
 - Hydropower from reservoirs has the greatest impact on water by far. Countries that rely on hydropower have a high WF
 - France's WF oscillates between 1.5L/kWh to 3.5L/kWh over the year...
- Join assessment of footprints
 - The results reveal that a shift to a decarbonized system can be detrimental to water resources if it mainly relies on hydropower from reservoirs, while other low-carbon intensive energy sources like solar and wind do not have an impact on water systems
 - Optimising both the CF and the WF at the same time can be challenging - as demonstrated by the case of Spain
 - The case of reservoir-based hydropower must be given particular attention when assessing the CF and WF of electricity generation. In contrast to most

other generation technologies, whose operational environment impacts are directly linked to electricity production, hydropower reservoirs exert a continuous impact regardless of whether electricity is being generated or consumed

- While demand-side strategies, such as shifting electricity consumption to periods of lower arginal CF and WF, can be effective in reducing the overall environmental impact of electricity systems, they are less effective in systems with a high share of reservoir-based hydropower.
- Closed-loop systems reuse the same water, reducing new water input but still facing evaporation losses from both reservoir, while open-loop system continuously draw water.
- Considering the importance of PHS reservoirs as a storage technology for power systems, and even more in a context where renewable energy sources have an increasing weight in the countries' energy mix, more thorough evaluations of their WF are needed

Conclusions:

- Dependence on digital services and the widespread use of artificial intelligence are among the factors causing electricity consumption to grow at an unprecedented rate
- Having access to precise information about the impact of our electricity consumption is the first step to being conscious about this and then to take action
- Wattnet: open-source tool for the joint assessment of the CF and WF of electricity consumption, explicitly accounting for the spatial and temporal dynamics of electricity generation and cross-border energy flows in Europe
- The results demonstrate that ignoring electricity imports and exports can lead to systematic under- or overestimation of both CF and WF. Countries that are net importers of electricity or that rely on electricity from hydropower-intensive regions are particularly affected.
- Likewise, the use of monthly or annual average emission factors conceals short-term variability that is critical for demand-side strategies such as **workload shifting in DCs**
- Reservoir-based hydropower can significantly increase WF despite its low operational carbon intensity - consequently, carbon-centric optimization strategies may inadvertently exacerbate water stress if water impacts are not considered explicitly.
- The findings evidence the need for integrated carbon-water assessments.
- Beyond digital infrastructures, the methodology applies to any electricity-intensive activity where minimizing both carbon emissions and water use is a relevant sustainability objective

Metrics:

- The power usage effectiveness (**PUE**) is a widely used metric for energy usage efficiency.
- **Water Use Effectiveness (WUE)** is used to assess the water usage per unit of IT energy consumption. It is similar to the concept of the WF, which measures the amount of water used in a particular activity, but applied to DC's operation

Data Center water consumption (2021) (cited by gone with the clouds, uses PUE like in wattnet)

- To reliably serve these billions of users, internet properties rely on millions of dedicated computers called servers. These servers are located in data centers which provide reliable power, cooling and internet access.
- A less well understood factor is water consumption
- Projections suggest that water demand will increase by 55% between 2000 and 2050 due to growth from manufacturing (+400%), thermal power generation (+140%) and domestic use (+130%).
- Google reported 15.8 billion liters of water consumption, up from 11.4 billion liters...
- Paper: examines the water consumption of data centers, how that consumption is measured by the ICT sector, and considers where the industry is going in attempts to reduce water consumption
- In 2014, a total of 626 billion liters of water use was attributable to US data centers
- A medium-sized data center uses as much water as three average-sized hospitals or more than two 18-hole golf courses
- Some data center operators are drawing more than half of their water from potable sources - this has led to controversy in areas of water stress and highlights the importance of understanding how data centers use water
- Power generation source is therefore a key element in understand data center water consumption, with PUE as an indicator of how efficiently that power is used

Water use in data centre cooling:

- Drawing cool air over hot metal transfers heat energy to that air, which is then pushed out into the environment. The same process occurs in data centers, just at a larger scale. ICT equipment is located within a room or hall, heat is ejected from the equipment via an exhaust and that air is then extracted, cooled and recirculated.
- As temperatures increase, equipment failure also increases
- In cooler regions, less cooling is required and instead free air cooling can draw in cold air from the external environment

Measuring data center water use:

- $WUE = \text{Annual site water usage} / \text{ICT equipment energy}$
 - This offers a limited view because it only includes the water consumed on-site
 - water from electricity generation is also important for understanding the total data centre footprint
- $WUE \text{ source} = (\text{Annual source energy water usage} + \text{annual site water use}) / \text{ICT equipment energy}$
- This may be one reason why less than a third of data centre operators track any water metrics and water conservation is ranked as a low priority

Cloud vendor water use:

- Cloud computing is a major reason behind claims about flattening data centre energy usage over the last few years

Most important stuff:

- Data centers consume water across two main categories:
 - Indirectly through electricity generation (traditionally, thermoelectric power)
 - Directly through cooling
- Water requirements are measured based on withdrawal or consumption
 - Consumption: water loss (usually evaporation)
 - Withdrawal: water taken from a source such as natural surface water, underground water, reclaimed water or treatable potable water and then alter returned to the source
- Fuel and generator technology type have a major impact on cooling water requirements:
 - Dry air cooling system
 - Wet cooling system
- Hydropower systems also use large volumes of water despite being considered a cleaner source of electricity. Water evaporation from open reservoirs is a major source of losses, particularly in dry regions where water is not pumped back into the reservoir or passed onto downstream users
- PUE is a measure of how much of energy input is used by the ICT equipment as opposed to the data center infrastructure such as cooling:
 - $PUE = \text{Data center total energy consumption} / \text{ICT equipment energy consumption}$
 - PUE is relevant to understanding indirect water consumption
 - An ideal PUE of 1.0 would mean 100% of the energy going to power useful services running on the ICT equipment rather than wasted on cooling, lighting and power distribution. Water is consumed indirectly through the power generation
 - The largest “hyperscale” cloud providers report PUEs ranging from 1.25 to 1.8
- Direct water use is also important - all that equipment needs cooling, which in some older facilities can consume up to 30% of total data centre energy demand
- WUE: Facebook publishes real-time online dashboards for their Luleå, Forest City and Princeton data centers

Adaptive, Efficient, and Fair Resource Allocation in Cloud Datacenters leveraging Weighted A3C Deep Reinforcement Learning

Energy, exergy and computing efficiency based data center workload and cooling management (cited by Google with the clouds)

Papers of workload executions (5,6 cited by WattNet)

Structure of introduction:

Facts and why more focus should be on water

Previous implementations

Research Questions:

Relevant Work:

Previous research

Metrics

Dataset descriptions